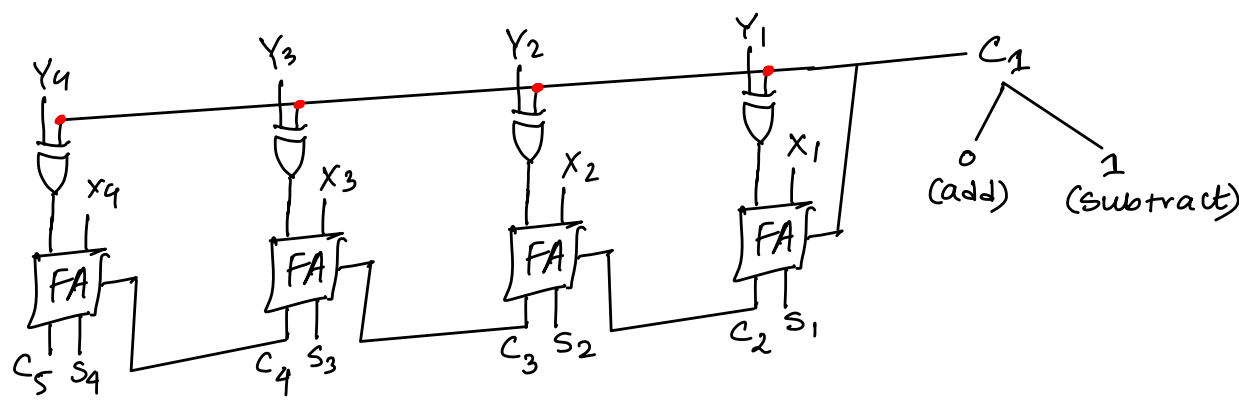
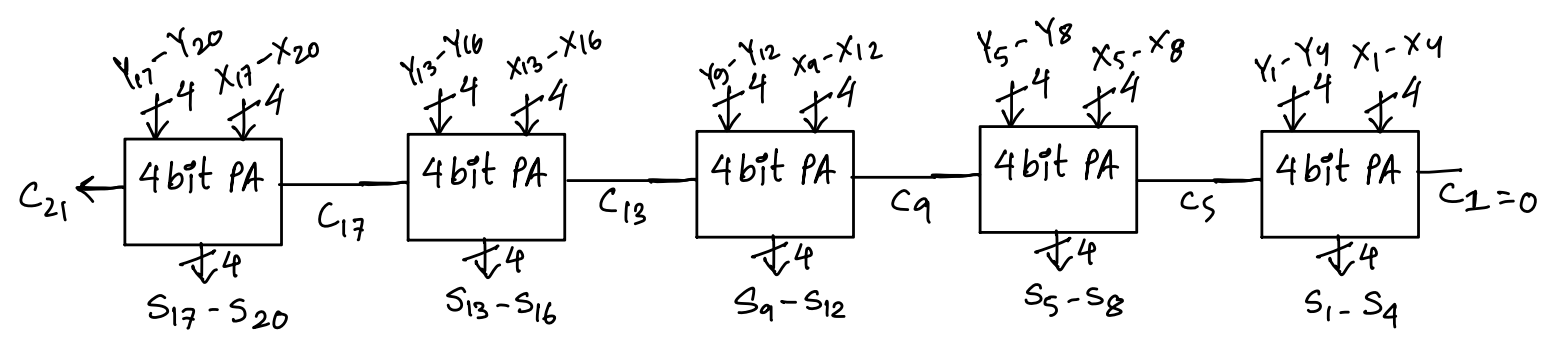


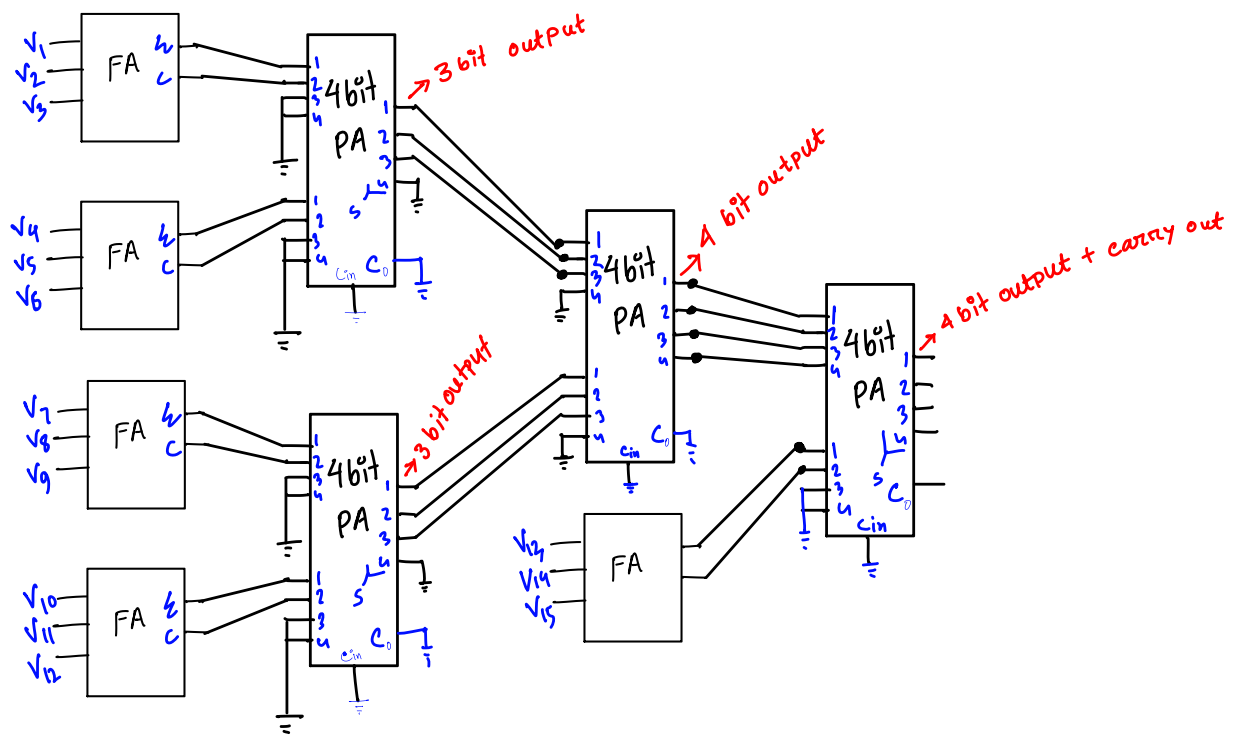
⇒ Build an adder cum subtractor (4 bit)



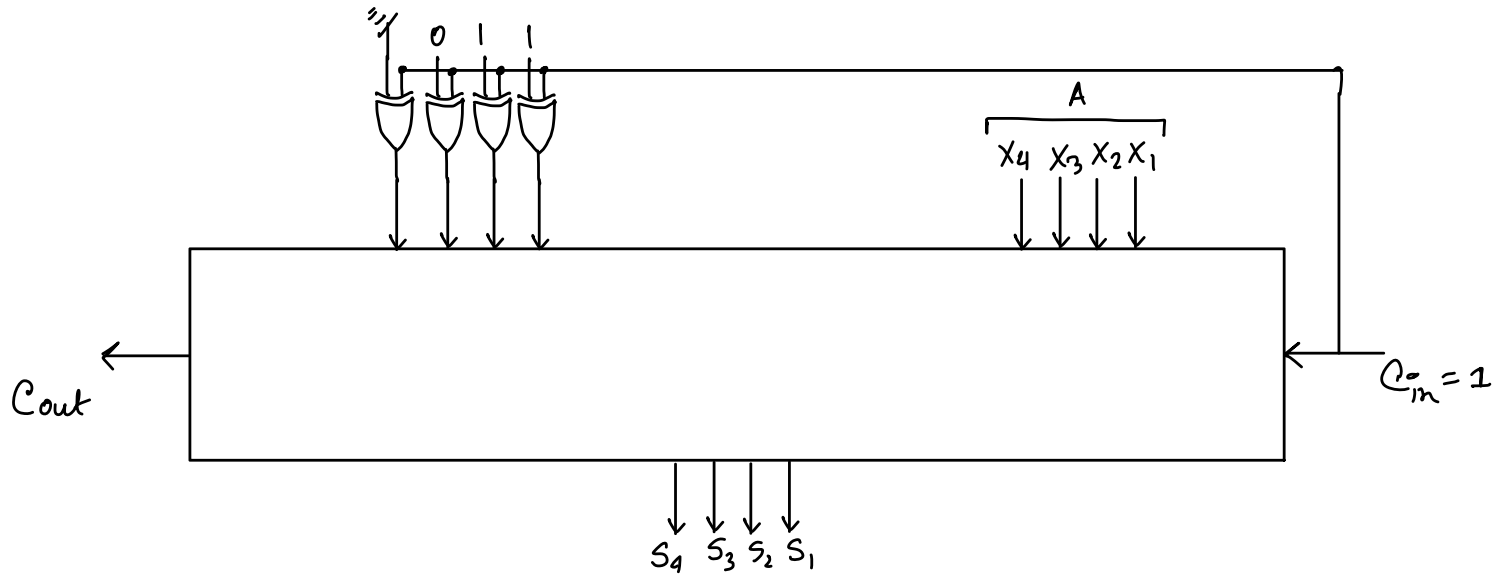
⇒ Block Diagram of a 20 bits parallel adder



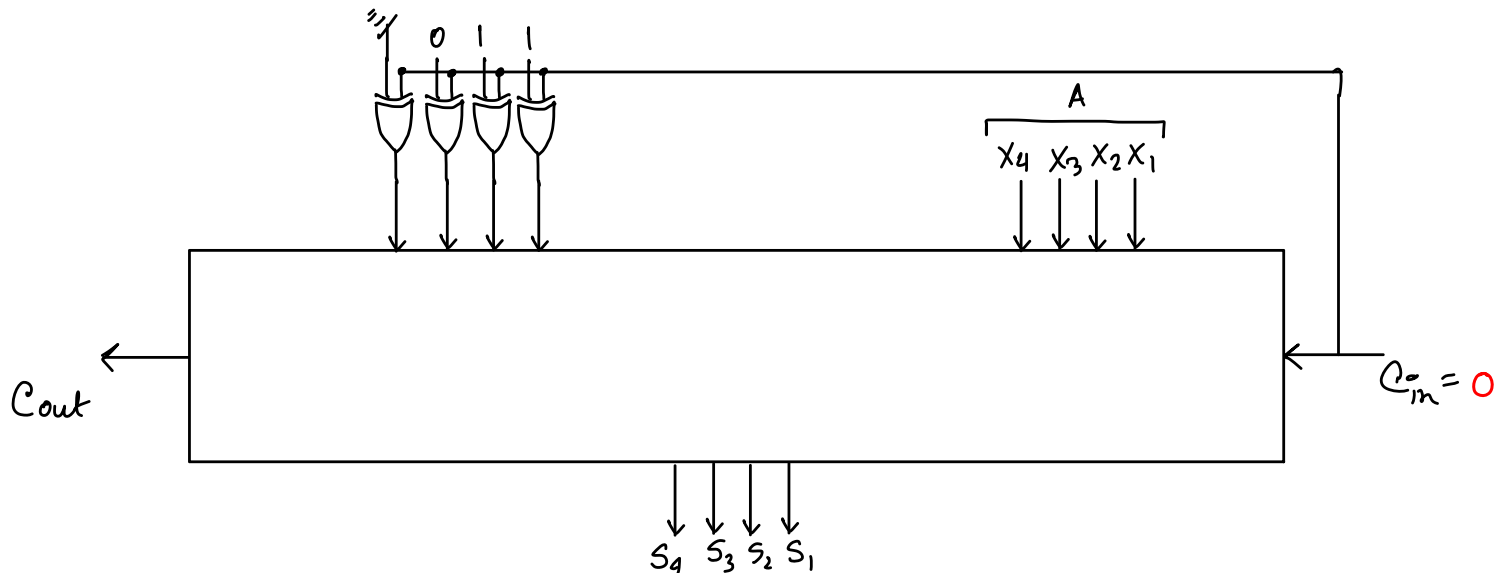
⇒ Build a 15 person voting system using FA and PA



⇒ Consider A is a 4 bit number. Design A-3 using a 4 bit parallel adder. Use external gate if required



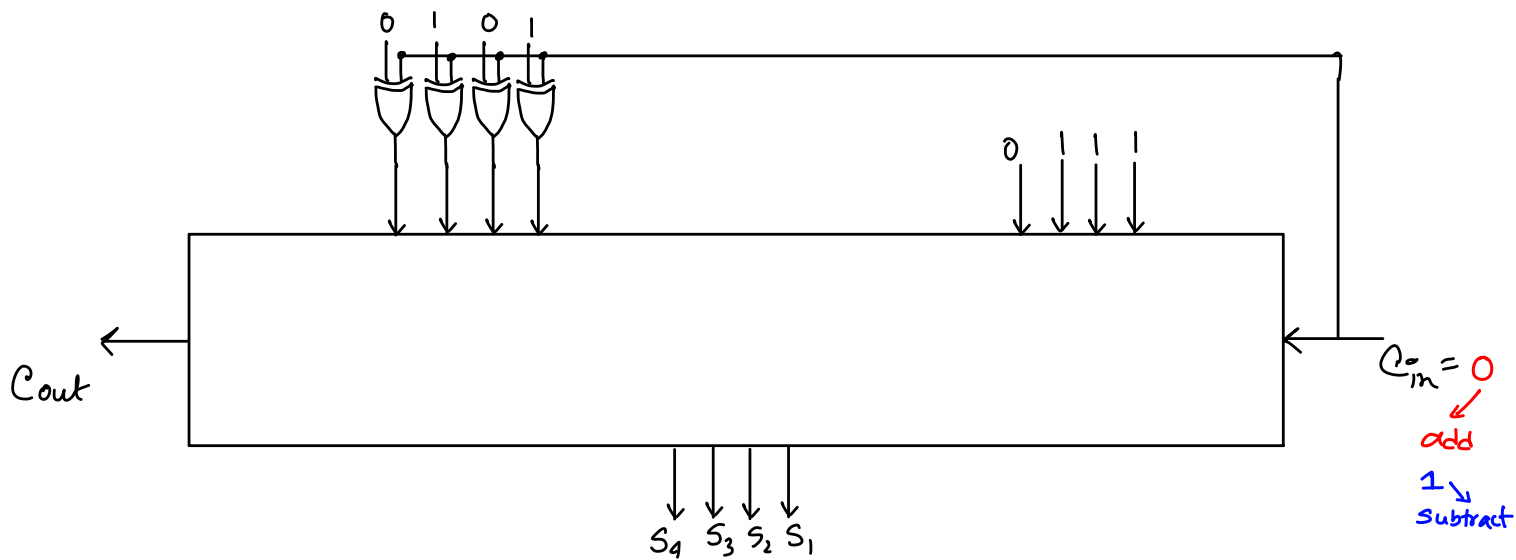
⇒ Consider A is a 4 bit number. Design $A+3$ using a 4 bit parallel adder. Use external gate if required



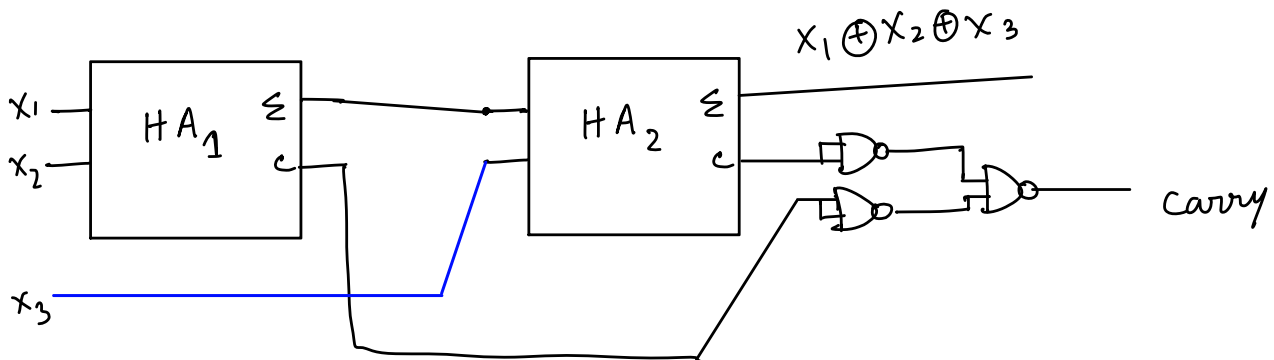
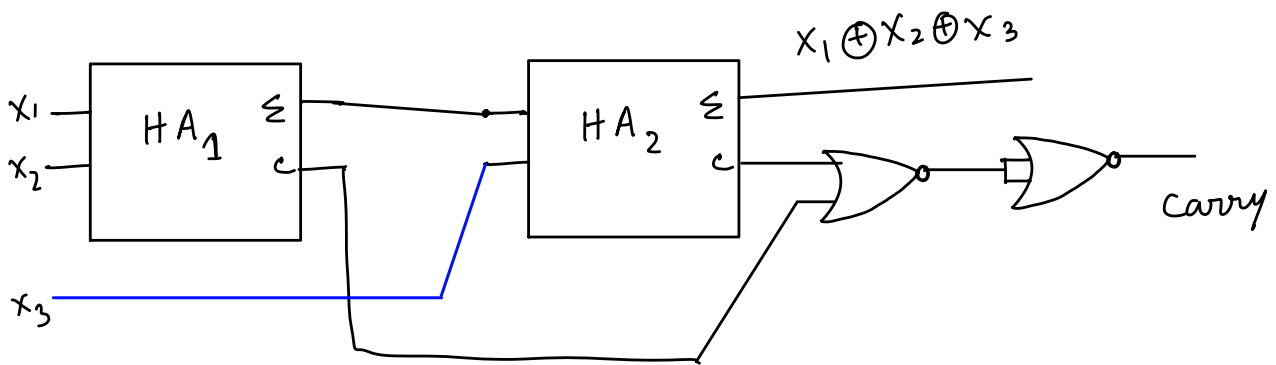
- Consider two numbers: 7 and 5. You can only calculate addition and subtraction between those two numbers. Design a circuit that can perform the above calculations based upon the user's intention.

$$7 = 0111$$

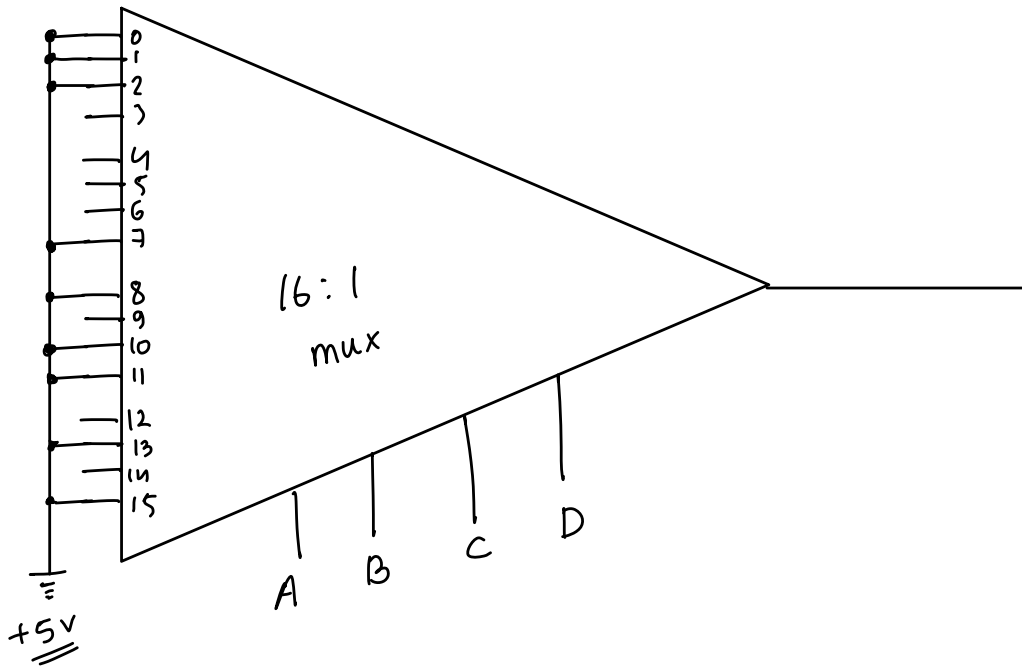
$$5 = 0101$$



- Design a full adder using two half adders. You must use two NOR gates and no OR gates.
- Design a full adder using two half adders. You must use three NAND gates and no OR gates.



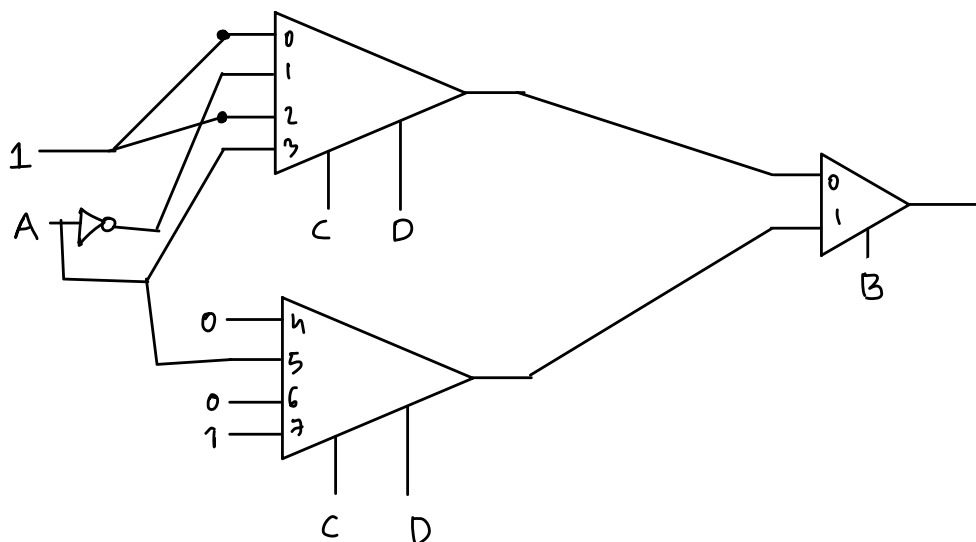
- Implement the following boolean function using a single 16:1 mux .
 $F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$. Use external gates if required.



- Implement the following boolean function using both 4:1 and 2:1 mux in a single circuit
 $F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$. Use external gates if required.

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	
A'	0	1	2	3	4	5	6	7	
A	8	9	10	11	12	13	14	15	
	1	A'	1	A	0	A	0	1	

A	B	C	D	F	
0	0	0	0	1	0
0	0	0	1	1	1
0	0	1	0	1	2
0	0	1	1	0	3
0	1	0	0	0	4
0	1	0	1	0	5
0	1	1	0	0	6
0	1	1	1	1	7
1	0	0	0	1	8
1	0	0	1	0	9
1	0	1	0	1	10
1	0	1	1	1	11
1	1	0	0	0	12
1	1	0	1	1	13
1	1	1	0	0	14
1	1	1	1	1	15

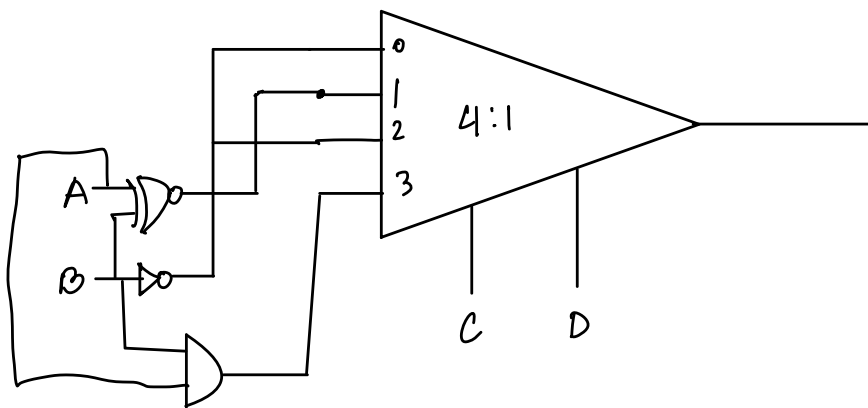


- Implement the following boolean function using a single 4:1 mux.
 $F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$. Use external gates if required.

	I_0	I_1	I_2	I_3
$A'B'$	1 ₀	1 ₁	1 ₂	
$A'B$				1 ₇
AB'	1 ₈		1 ₁₀	1 ₁₁
AB		1 ₁₃		1 ₁₅
	B'	$A \oplus B$	B'	$A+B$

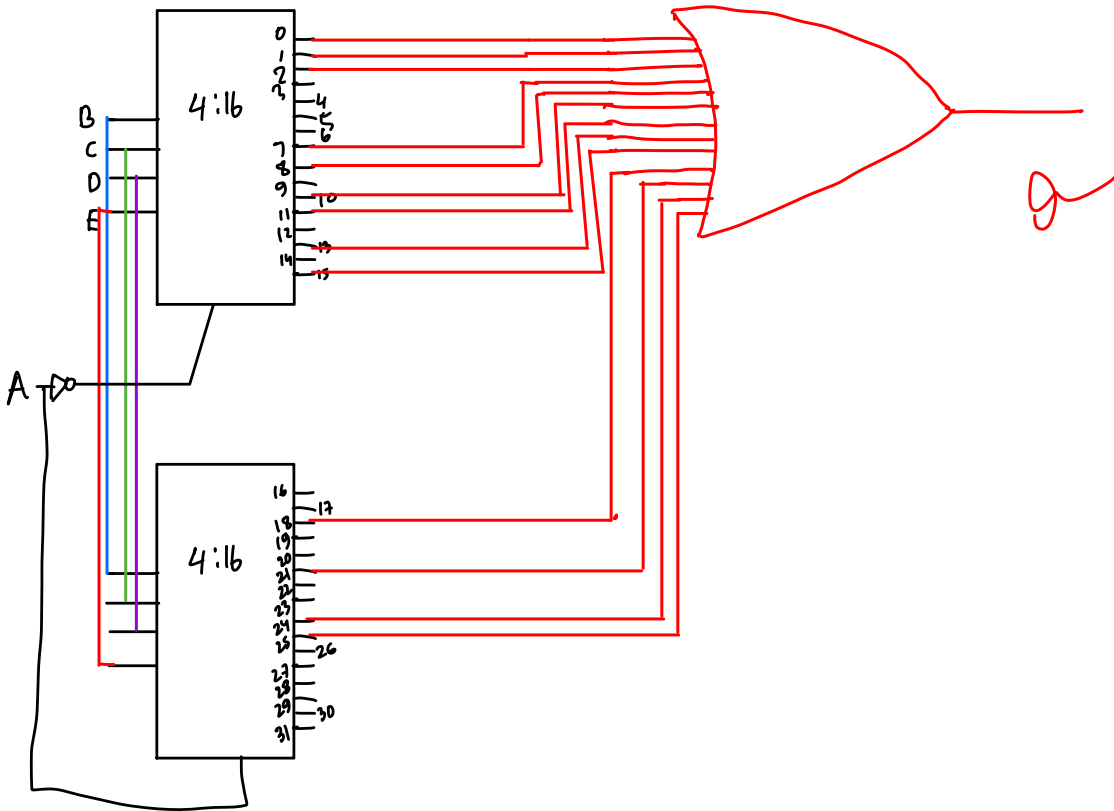
$$\begin{aligned}
 &A'B + AB' + AB \\
 &= A'B + A(B' + B) \\
 &= A'B + A \cdot 1 \\
 &= A + B
 \end{aligned}$$

A	B	C	D	F	
0	0	0	0	1	0
0	0	0	1	1	1
0	0	1	0	1	2
0	0	1	1	0	3
0	1	0	0	0	4
0	1	0	1	0	5
0	1	1	0	0	6
0	1	1	1	1	7
1	0	0	0	1	8
1	0	0	1	0	9
1	0	1	0	1	10
1	0	1	1	1	11
1	1	0	0	0	12
1	1	0	1	1	13
1	1	1	0	0	14
1	1	1	1	1	15

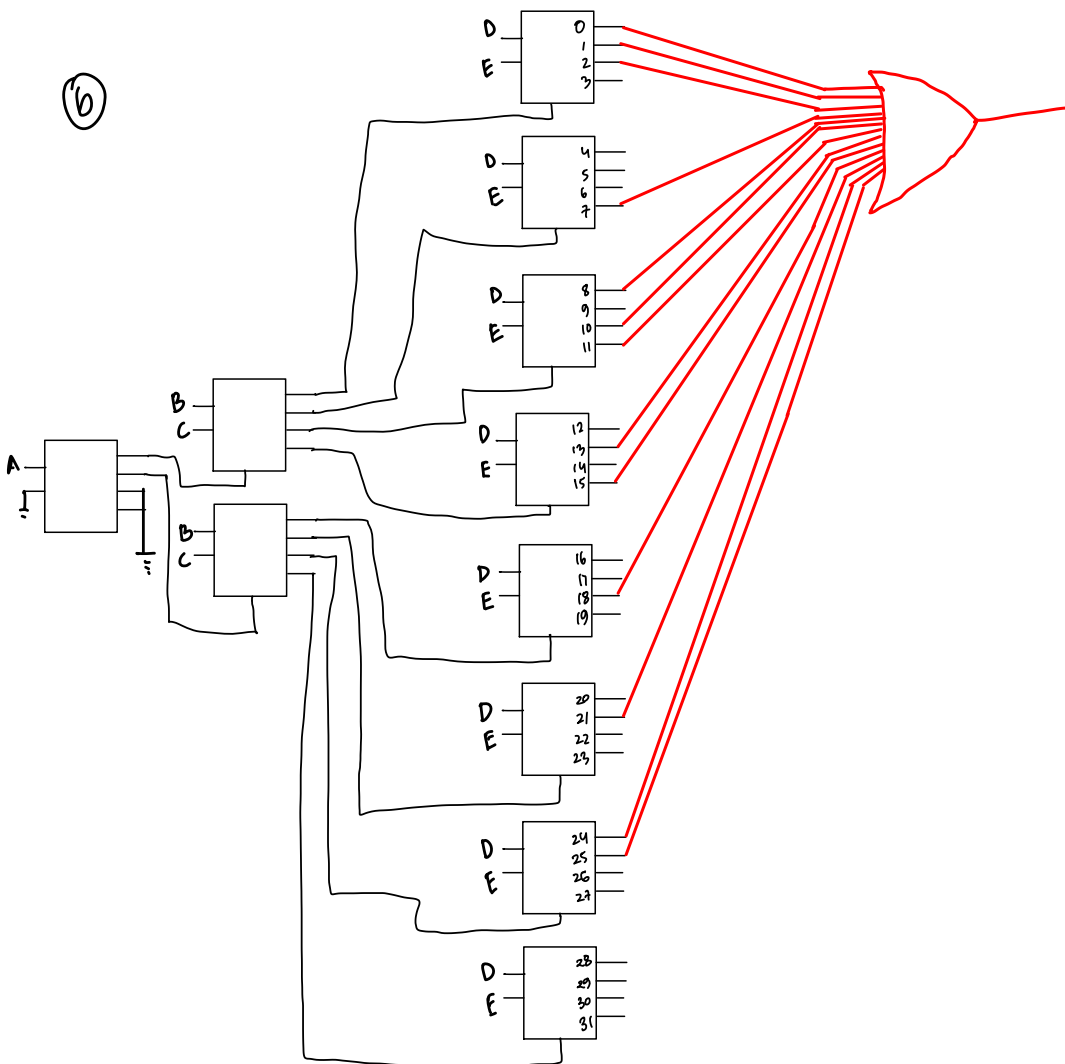


- Implement the following boolean function using **a) 4x16 decoder(s) only** **b) 2x4 decoder(s) only**
 $F(A,B,C,D,E) = \sum(0,1,2,7,8,10,11,13, 15,18,21,24,25)$. Use external gates if required.

②



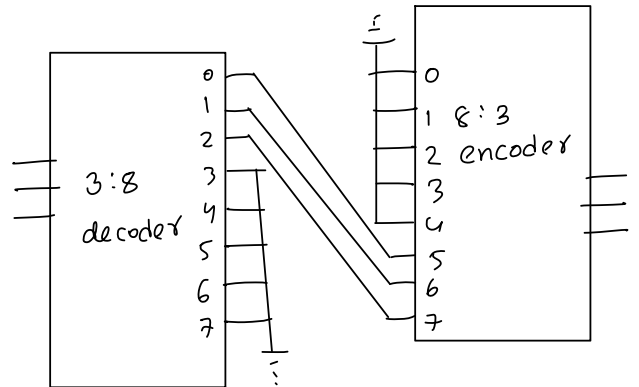
⑥



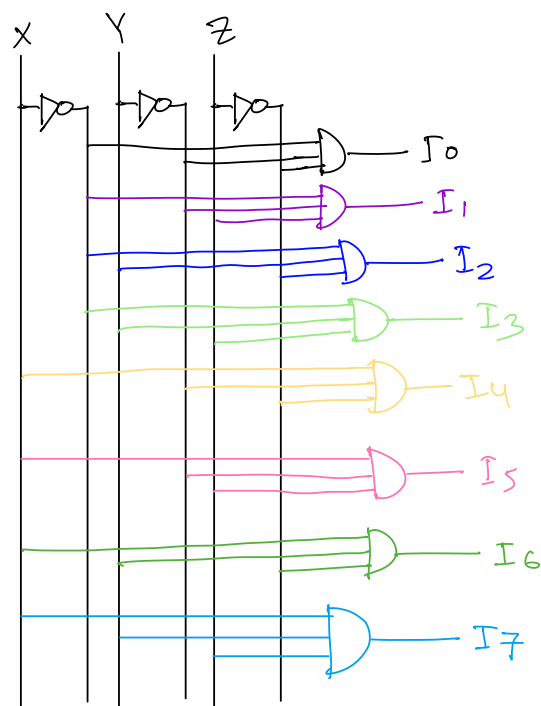
- Build a BCD to Excess-5 code converter using encoder(s) and decoder(s).

0	0	0		1	0	1
0	0	1		1	1	0
0	1	0		1	1	0
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

$0 \rightarrow 5$
 $1 \rightarrow 6$
 $2 \rightarrow 7$

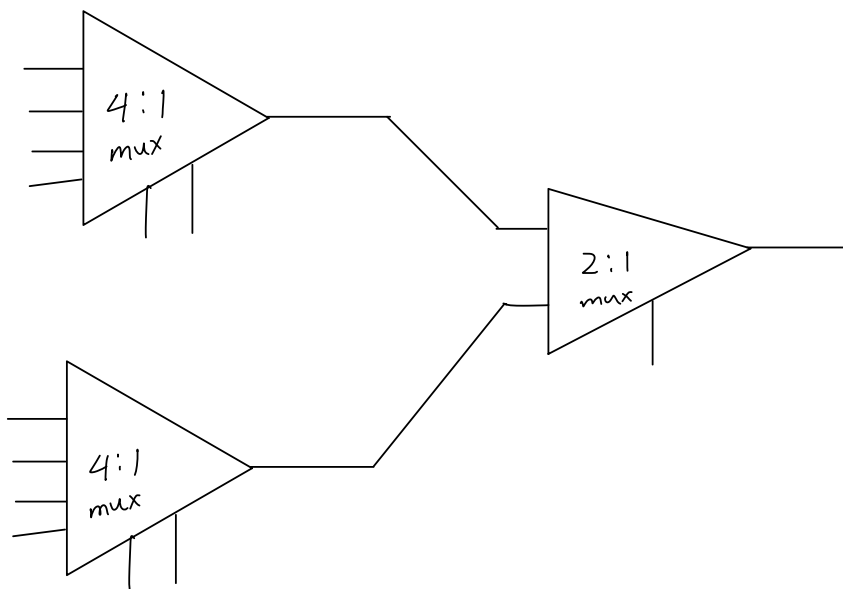


- Design the circuit diagram for a 3x8 decoder.

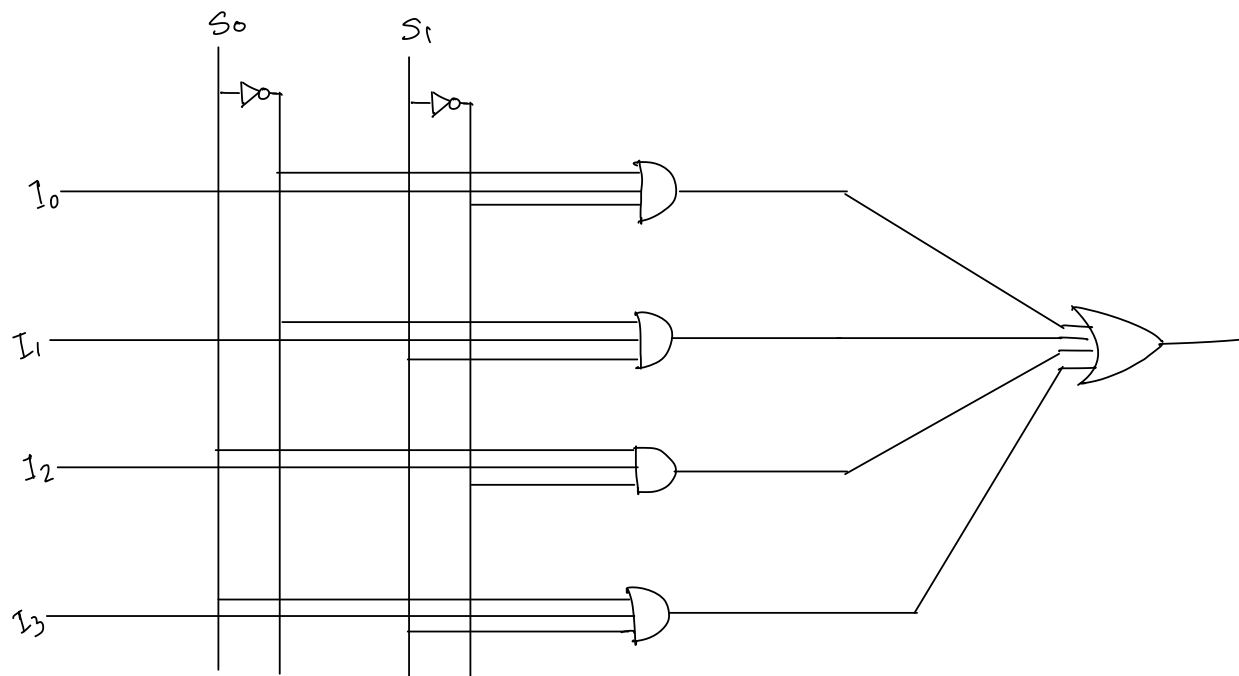


X	Y	Z	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	
0	0	0	1								$x'y'z'$
0	0	1		1							$x'y'z$
0	1	0			1						$x'yz'$
0	1	1				1					$x'yz$
1	0	0					1				$xy'z'$
1	0	1						1			$xy'z$
1	1	0							1		xyz'
1	1	1								1	xyz

- Design a 8:1 mux using both 4:1 mux and 2:1 mux in a single circuit.



- Design AND, OR, NOT gate using 4:1 mux.



- Design AND, OR, NOT gate using 2:1 mux.

